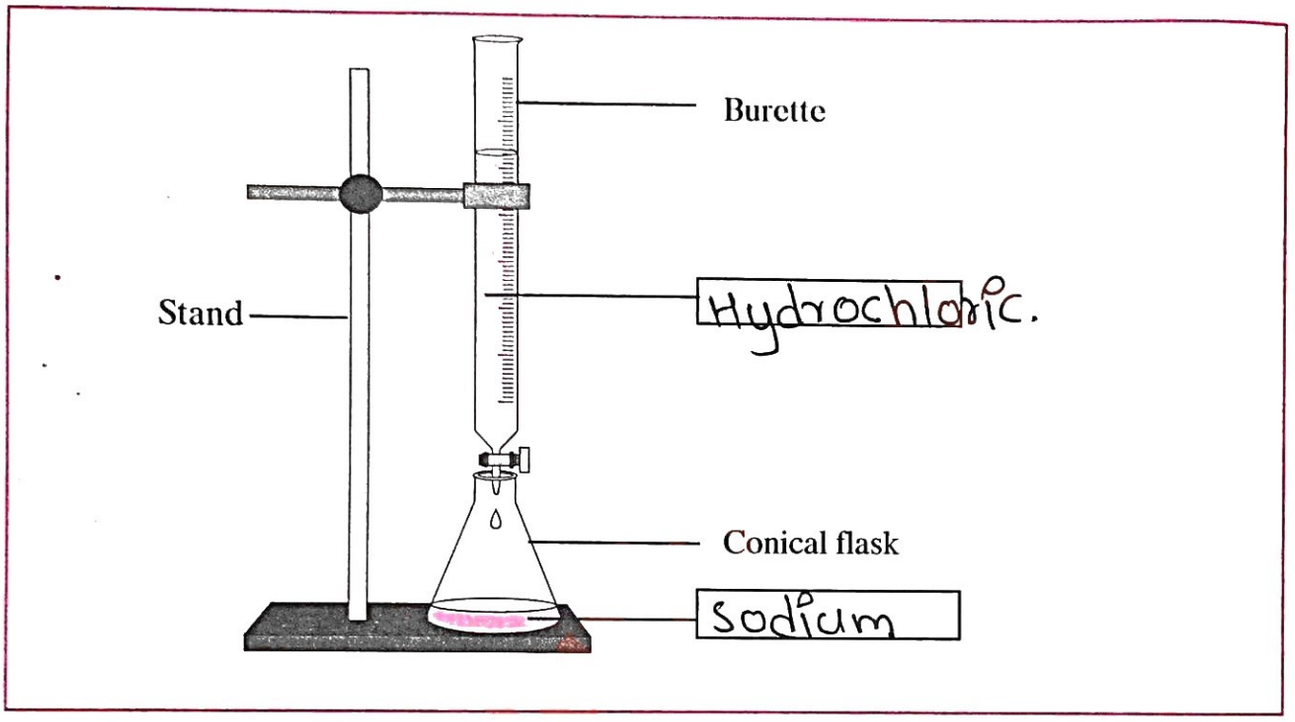
Experiment No.6

Title : To study the reactivity of Acid and Base in terms of Neutralization principle.

Apparatus : 2 beakers, Burette, Pipette, Conical flask, Clean white tile, Stand etc.

Chemicals : 0.1 N Hydrochloric acid, 0.1 N Sodium hydroxide, Phenolpthalein Indicator, etc.

Figure :



Procedure :

1. Fix a burette on a stand and fill it with dilute hydrochloric acid upto zero mark. (Take a precaution that there is no air bubble, by taking out the solution rapidly through the stopcork before filling the solution upto the '0' mark.)
2. Pipette out 10 ml NaOH solution in conical flask.
3. Put 2 to 4 drops of phenolphthalein in the conical flask.
4. The solution in the conical flask will turn pink.
5. Keep the conical flask on the tile and release 1 ml HCl at a time in it from the burette.
6. Swirl the conical flask. Do this procedure until the solution in the conical flask becomes colourless.
7. Note the reading as pilot reading on turning the solution colourless .
8. Repeat the steps 1 to 4.
9. Release the solution from the burette upto the pilot reading. Swirl the conical flask.
10. Now add the solution from burette dropwise into a conical flask carefully. Swirl the conical flask continuously.
11. Note the burette reading as soon as the solution in the conical flask turns colourless.
12. Take three such readings by changing the solution in the conical flask and note these in the observation table.

Observation :

1. Burette Solution ..Hydrochloric acid.
2. Solution in the Conical flask ..Sodium hydroxide.
3. Indicator .phenaptatein.
4. End Point : Colour of the solution in the conical flask – faint pink to .colourless.
5. Pilot Reading for completion of reaction : ..2.5....ml to ...3....ml
6. Chemical Reaction :

$$HCl + NaOH \rightarrow NaCl + H_2O.$$

Accurate Readings :

Burette Reading	Reading in ml			Constant Burette Reading in ml
	I	II	III	
Initial	0.0	0.0	0.0	0ml.
Final	3.5	3	3	3ml.
Difference	3.5	3	3	3ml.

Inference :

.....when acid base reacts with each other salts and water is formed.....

Multiple Choice Questions

1. Phenolphthalein indicator is a ..... indicator.
   A. Natural ✓B. Chemical C. Universal D. All of above
2. When 4 drops of phenolphthalein are added to hydrochloric acid, the solution is seen to be .....
   A. Red ✓B. Pink C. Yellow D. Colourless
3. The neutralization reaction takes palce in ..... and .....
   A. Acid, Acid ✓B. Acid, Base C. Base, Salt D. Base, Indicator

Questions

1. Which measures are taken to decrease acidity in body? Why?
   .....we should take have such as an acid tabile or milk of the magnesia, it neutrases the effect of an Acid.....
2. How are the basic salts are formed?
   .....when a strong base reacts with a weak acid base salt are formed.....

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